SMARTPAY

DIGITAL PAYMENT APPLICATION

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1. Project Title

SmartPay – Digital Payment Application

2. Introduction

Digital payment applications provide users with a convenient way to manage money and perform everyday payment activities. SmartPay is an Android-based digital payment application designed to demonstrate the basic functionality of a digital payment system.

The application allows users to register and log in, manage a digital wallet, add money, send money to another registered user, make QR-based payments, and view transaction history. The application provides a simple and user-friendly interface where users can easily access different payment features.

Firebase Authentication is used for user registration and login, while Firebase Realtime Database is used to store user details, wallet balances, and transaction information. The payment functionality is designed as a demo/simulated payment system for academic purposes. The project focuses on Android, Java, XML, and Firebase integration.

3. Problem Statement

- Managing different payment activities can be inconvenient without a simple interface.
- Users need an easy way to view and manage their wallet balance.
- Sending money between users should be simple and easy to understand.
- Maintaining previous transaction details manually can be difficult.
- There is a need for a simple application that demonstrates the basic working of a digital payment system.
- Existing payment applications often include features that are unnecessary for demonstrating basic payment functionality.

4. Objectives

- Allow users to register and log in to the application.
- Provide a digital wallet for users.
- Allow users to add money to their wallet.
- Allow users to send money to another registered user.
- Provide QR-code-based payment functionality.
- Display the current wallet balance.
- Maintain a record of previous transactions.
- Display payment confirmation and transaction status.
- Provide a simple user profile.
- Store user and transaction information using Firebase.
- Provide a simple and user-friendly Android application.

5. Scope

- The application is designed for students and general users.
- Users can register and log in to their SmartPay account.
- Users can maintain a demo digital wallet.
- Users can add a specified amount to their wallet.
- Users can send money to another registered SmartPay user.
- Users can make payments using a QR code.
- Users can view their previous transactions.
- Users can view their basic profile information.
- Firebase can be used for storing user and transaction data.
- The system can be extended with additional payment features in the future.

Note: The payment system will be a simulated/demo payment system and will not process real bank, UPI, or other financial transactions.

6. Modules of the System

Module 1 – User Authentication Module

Purpose: To allow users to securely access the SmartPay application through registration and login.

Key Features:

- Firebase Authentication with email/password login.
- New user registration.
- Existing user login.
- User logout.
- Basic user profile creation.
- Secure authentication using Firebase Auth.

Module 2 – Digital Wallet Module

Purpose: To allow users to manage their demo wallet balance.

Key Features:

- Display current wallet balance.
- Add money to wallet.
- Update wallet balance.
- Store wallet balance in Firebase Realtime Database.
- Display wallet information on the home screen.

Module 3 – Send Money Module

Purpose: To allow users to transfer demo money to another registered SmartPay user.

Key Features:

- Enter receiver details.
- Enter payment amount.
- Check available wallet balance.
- Confirm payment.
- Deduct amount from sender's wallet.
- Add amount to receiver's wallet.
- Display payment confirmation.
- Store transaction details.

Module 4 – QR Payment Module

Purpose: To provide a simple QR-code-based payment facility.

Key Features:

- Scan QR code.
- Identify the receiver.
- Enter payment amount.
- Confirm payment.
- Update wallet balance.
- Store payment details.
- Display payment status.

Module 5 – Transaction History Module

Purpose: To allow users to view their previous payment activities.

Key Features:

- Display transaction history.

- Show transaction amount.
- Show sender and receiver.
- Display transaction type.
- Display date and time.
- Display transaction status.
- Store transaction records in Firebase.

Module 6 – Profile Module

Purpose: To allow users to view their basic account information.

Key Features:

- Display user name.
- Display registered email.
- Display phone number.
- View basic account details.
- Logout option.

7. Technology Stack

Layer	Technology Used
Frontend	Java
User Interface	XML
Development Tool	Android Studio
Platform	Android
Authentication	Firebase Authentication
Database	Firebase Realtime Database
QR Payment	QR Code Scanner
Programming Language	Java

Firebase Authentication will be used for registration, login, and logout. Firebase Realtime Database will be used for user details, wallet balance, transactions, and QR payment information.

8. Database Structure

The application will use Firebase Realtime Database for storing application data.

Main database nodes:

SmartPay Database
├── users
├── transactions
└── QRPayments

Users

users → USER_ID → name, email, phone, balance.

Transactions

transactions → TRANSACTION_ID → senderId, receiverId, amount, type, status, timestamp

QR Payments

QRPayments → QR_ID → userId, userName

9. Hardware & Software Requirements

Hardware:

- Android smartphone for testing.
- Laptop/PC with minimum 4 GB RAM.
- Internet connection.

Software:

- Android Studio.
- Java JDK.
- Android SDK.
- Firebase Console.
- Firebase Authentication.
- Firebase Realtime Database.
- QR Code Scanner library.
- Internet connection.

10. Application Structure

SmartPay
├── Login
├── Register
├── Home
│ ├── Wallet Balance
│ ├── Add Money
│ ├── Send Money
│ ├── QR Pay
│ ├── Transactions
│ └── Profile
├── Add Money
├── Send Money
├── QR Payment
├── Transaction History
└── Profile

11. Expected Outcome

- User-friendly Android application for basic digital payments.
- Secure user registration and login.
- Digital wallet for managing a demo balance.
- Ability to add money to the wallet.
- Ability to send money between registered users.
- QR-based payment functionality.
- Wallet balance management.
- Transaction history.
- Simple user profile.
- Firebase-based storage of user and transaction information.
- Simple and easy-to-use interface.

12. Conclusion

SmartPay – Digital Payment Application provides an Android-based platform for demonstrating the basic functionality of a digital payment application. The application allows users to register and log in, manage a demo wallet, add money, send money to other registered users, make QR-based payments, and view their transaction history.

The system uses Java and XML in Android Studio, with Firebase Authentication for user login and registration and Firebase Realtime Database for storing user, wallet, and transaction information. The project is intentionally designed to be suitable for a college-level Android project and does not process real financial transactions.

Overall, SmartPay provides a practical and easy-to-understand implementation of a digital payment application using Android and Firebase, while leaving scope for additional features in future versions.